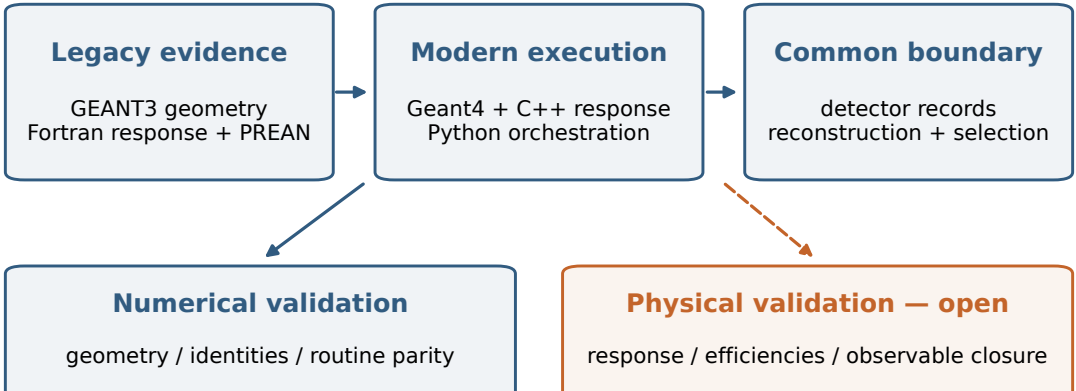
Legacy evidence
GEANT3 geometry
Fortran response + PREAN
Modern execution
Geant4 + C++ response
Python orchestration
Common boundary
detector records
reconstruction + selection
Numerical validation
geometry / identities / routine parity
Physical validation — open
response / efficiencies / observable closure